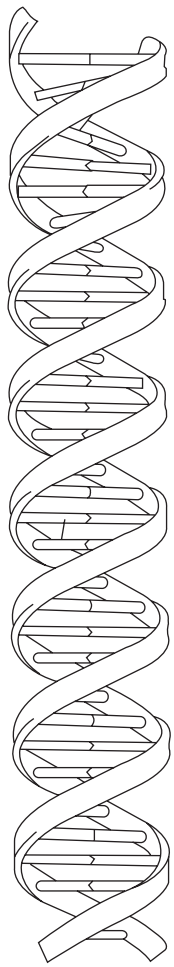
DNA discoveries

DNA stands for deoxyribonucleic acid. It is found in the **nuclei** of cells and contains genetic material called **genes**. Since the 1940s scientists have known that DNA is the **molecule** of life, but they didn't know what the molecule looked like. They knew DNA consisted of four **bases**: adenine (A), thymine (T), guanine (G) and cytosine (C). These bases joined together in pairs; A always pairs with T, and G always pairs with C. Scientists also knew that the structure had a phosphate backbone but they weren't sure how the bases formed along it. In 1953 two scientists named Watson and Crick proposed that the bases are paired like rungs in a twisted ladder that attaches to the phosphate backbone. The formation is called a **double helix**.

☞ **1** Read the passage above and underline any information about the structure of a DNA molecule.

2 Use the word bank to complete the sentence describing the illustration.

Word bank				
bases	outside	phosphate	molecule	four



The p_____ backbone is on the o_____ and the f_____ different b_____ are on the inside of the DNA m_____.

3 Decide whether these statements are true or false.

a) Watson and Crick were the only scientists working on DNA. _____

b) Watson and Crick discovered the DNA double helix. _____

c) DNA has four bases: A, T, G and C. _____

d) The phosphate backbone is on the inside of the DNA molecule. _____

e) G and T always pair up together. _____

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